Technical Project Report

Cynbit Technologies – Website Enhancement with AI-Powered Company Chatbot

Project Type: Website Enhancement & AI Integration

Organization: Cynbit Technologies

Prepared by: [Your Name]

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1. Introduction

This project focuses on improving the digital presence of Cynbit Technologies by developing an enhanced, modern, and interactive version of its existing website.

The primary objective was to understand the company's business, services, and offerings, and develop a website that represents its identity while improving user engagement through an AI-powered chatbot.

The chatbot acts as a virtual company representative, allowing visitors to ask questions about the organization, its services, and its offerings through a conversational interface.

2. Problem Statement

A company website is an important communication channel between an organization and its potential customers. However, visitors may need to navigate multiple sections to find specific information about services, products, and company offerings.

The objective of this project was to:

• Develop a modern and responsive website representing Cynbit Technologies.

• Improve the presentation and accessibility of company information.

• Integrate an AI-powered chatbot for interactive communication.

• Restrict chatbot responses to company-related queries.

• Provide API configuration flexibility by allowing users to select an AI provider and supply their API key.

3. Company Research and Understanding

Before development, the existing Cynbit Technologies website was studied to understand its business, services, and organizational identity.

Cynbit Technologies is a technology company based in Jaipur, Rajasthan. Its areas of operation include AI automation, custom software development, web and mobile application development, cloud infrastructure, digital marketing, and SaaS products.

The website's content and structure were used as a reference for developing the enhanced version.

4. Proposed Solution

The proposed solution consists of two major components:

A. Enhanced Company Website — A modern, user-friendly website designed to communicate the company's identity, services, and offerings effectively.

B. AI-Powered Company Chatbot — An interactive chatbot integrated into the website to provide responses to visitors' questions about Cynbit Technologies. It is designed to behave like a company representative rather than a general-purpose AI assistant.

5. Technology Stack

Technology choices should reflect the actual implementation. Complete the bracketed entries below before submitting the report.

Component	Technology used
Frontend	[Enter actual framework/library]
Styling	[Enter actual CSS/styling framework]
Backend	[Enter actual backend technology, or N/A]

Component	Technology used
AI Integration	[Enter actual AI provider/API]
Development Environment	Visual Studio Code
Version Control	[Git/GitHub, if used]

Technology selection was intended to support maintainable interfaces, external AI integration, flexible configuration, and future enhancements.

6. System Implementation

6.1 Website Development

The website was developed by taking inspiration from the original Cynbit Technologies website. The implementation focused on a modern user interface, company introduction, presentation of services, navigation between sections, chatbot integration, and responsive design.

6.2 AI Chatbot Integration

An AI-powered chatbot was integrated into the website to improve visitor interaction. Its key functionalities include:

• Interactive chat interface through which users can submit questions.

• API key configuration, allowing users to select an AI provider and provide the corresponding API key.

• Company-specific instructions and information to guide responses about Cynbit Technologies.

• Natural-language responses to user questions.

• Restricted conversation scope: the chatbot is instructed to handle company-related questions and politely decline unrelated queries.

6.3 Chatbot Working Process

1. The user opens the chatbot.

2. The user configures the AI provider and API key.

3. The user submits a question.

4. The application processes the question with company-specific instructions and context.

5. The request is sent to the configured AI service.

6. The AI generates a response, which is displayed in the chat interface.

6.4 High-Level Architecture

User → Website Chat Interface → API Configuration → AI Service → Generated Response → User

The company-specific instructions help maintain the chatbot's intended scope. Prompt instructions alone cannot guarantee that every unrelated or inaccurate response will be blocked; stronger enforcement can be added through server-side validation and a curated knowledge base.

7. Testing and Results

The implemented website and chatbot were reviewed for their primary functionality and user experience. The following test cases describe the expected behaviour and should be marked with actual pass/fail results after verification.

Test case	Expected behaviour	Actual result
Website navigation	Navigate between sections	[Record result]
Chatbot interface	Opens and accepts messages	[Record result]

Test case	Expected behaviour	Actual result
API configuration	Selected AI provider can be configured	[Record result]
Company questions	Relevant company information returned	[Record result]
Unrelated questions	Polite scope restriction	[Record result]
Response display	AI responses appear in chat	[Record result]

The two UI issues documented in Section 8 were identified during review and should be addressed in a subsequent refinement.

8. Identified UI Issues

Issue 1: Some text is not visible

Observation: Some text elements are not clearly visible on the website.

Possible causes: Insufficient contrast between text and background, opacity, or unsuitable text styling.

Impact: Reduces readability and negatively affects the user experience.

Recommended fix: Review text and background colors, opacity, font size, and typography to ensure sufficient contrast.

Issue 2: Cursor does not change to a pointer on hover

Observation: The cursor does not change to a pointer when hovering over certain buttons.

Possible cause: Missing pointer cursor styling.

Impact: Users may not immediately recognize clickable elements.

Recommended fix: Apply cursor: pointer to interactive buttons and clickable elements. Example CSS:

button, .clickable { cursor: pointer; }

9. Suggestions for Improving the Original Website

Suggestion 1: Detailed Project Case Studies

The original website could include a dedicated portfolio section containing detailed case studies of completed projects. Each case study could explain the client's problem, the proposed solution, technologies used, screenshots or demonstrations, and measurable results.

Expected benefit: Builds credibility, demonstrates technical expertise, and helps potential clients understand the company's capabilities.

Suggestion 2: Interactive Product Demonstrations

The website could provide interactive demonstrations or short guided walkthroughs of its SaaS products and technology solutions. These could include live previews, short demonstration videos, feature comparisons, and direct trial or demo-request options.

Expected benefit: Helps visitors understand the company's products more effectively and encourages potential customers to explore or request a demonstration.

10. Conclusion

The project demonstrates the enhancement of a company website through modern web development practices and AI integration. The addition of an AI-powered company-specific chatbot provides an interactive way for visitors to access company information and understand its services.

The project also demonstrates the practical application of external AI APIs, configurable AI integrations, prompt engineering, and user-interface development. Further improvements in text visibility, cursor interactions, and overall UI

refinement can make the website more polished and user-friendly.

11. Future Scope

• Integration of a company knowledge base using Retrieval-Augmented Generation (RAG).

• Multilingual chatbot support.

• Lead collection and customer inquiry management.

• Chat history and conversation management.

• Improved API key security and server-side integration.

• Accessibility and responsive UI improvements.

• Analytics to understand visitor interactions and frequently asked questions.